

Darkness Reads as Safety: A Consistent Appearance-Induced Failure Mode in End-to-End Driving Policies

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Paper ID —

Abstract—End-to-end driving policies are evaluated on benchmarks whose images are, almost without exception, clear and well-lit. We ask what their internal representations do when only the appearance of a scene changes, and find a failure mode that is consistent across four architectures: a synthetic night rendering moves the representation along a direction that is systematically opposed to the policy’s own hazard-response direction, and the planned speed rises accordingly. In representation space the hazard direction and the braking direction form one family (34–46° apart, against a 7° reproducibility floor and a 90° random baseline), while the appearance direction sits past the random line on the far side (93–115°). Behaviourally, the same perturbation raises planned speed by up to 2.05 m/s — an order of magnitude more than deleting the lead vehicle from the image. Two controls sharpen the mechanism. First, the effect is carried by sensor noise, not by low luminance: darkening alone changes planned speed by -0.01 m/s, and adding noise of the magnitude a real night camera produces flips it to $+0.05$. Second, perturbing only the sky — 17.9% of the frame, with the driving corridor untouched to the pixel — still recovers 19% of the effect of removing the hazard itself, so the dependence is global rather than mediated by any local cue. We also report, from five pre-registered tests, that the projection onto these directions is not a reliable proxy for per-policy or per-event behaviour; four of the five are rejected. Direction readouts are widely used as behavioural proxies, and we give multi-angle evidence for where that use breaks down.

I. Introduction

Driving benchmarks are built from clear, well-lit imagery. A policy that scores well on them has been asked, almost exclusively, to act on scenes whose appearance is easy. What its internal representation does when only appearance changes—and whether that change reaches the planned trajectory—is not something a leaderboard number reports.

We study this directly. Starting from mined human braking events, we extract three directions in each policy’s intermediate representation: the hazard-response direction v_{faith} , obtained from paired frames that differ only in whether the hazard is present; the braking direction v_{decel} , obtained by contrasting cruising and braking populations at matched speed; and the appearance direction v_{bright} , obtained by re-rendering the same frame as night. We then ask how these sit relative to one another, and what happens to the planned trajectory.

a) Finding.: The three directions do not spread out evenly. v_{faith} and v_{decel} cluster (34–46°; Table I); v_{bright}

sits on the opposite side of the random baseline from both. Because $v_{\text{faith}} = h(\text{hazard present}) - h(\text{hazard removed})$, a negative projection means the model reads “it got dark” the way it reads “the hazard went away.” The behavioural readout agrees: synthetic night raises planned speed by up to 2.05 m/s, where deleting the lead vehicle from the image raises it by 0.23 m/s. Two independent readouts—representation geometry and output trajectory—point to the same mechanism, in four architectures.

b) What carries it.: Darkness alone is nearly inert; the effect is carried by sensor noise. Perturbing only the sky, with the driving corridor untouched, still recovers a fifth of the hazard-removal effect. The dependence is therefore global, not mediated by a local cue in the region that matters for driving.

c) What we could not show.: A natural next step is to use the projection onto these directions as a cheap behavioural proxy—to rank policies, or to flag individual events, without closed-loop simulation. We pre-registered five such tests. Four are rejected, and the fifth holds for only two of four policies (Table III). We report this in full: the practice of treating a representation-space projection as a behavioural proxy is common, and our evidence says it does not hold up here. One readout does transfer across the domain shift—a multi-layer speed axis (Table IV)—and it is instructive that the one that works has an external target variable.

d) Contributions.:

- 1) An appearance-induced failure mode, consistent across four end-to-end policies, in which a night rendering moves the representation opposite to the policy’s own hazard direction and raises planned speed (Sections IV-B and IV-C).
- 2) Isolation of its cause: sensor noise rather than luminance, and a global rather than local dependence—established by controls that hold the driving corridor fixed (Section IV-D).
- 3) Five pre-registered tests of direction projections as behavioural proxies, four rejected, with the discipline that made the rejections interpretable (Section IV-E).

II. Related Work

a) Open-loop benchmarks and their discontents.: nuScenes [26] and the nuPlan-derived NAVSIM protocol [25] score a policy by comparing its plan against logged human behaviour, the latter through a non-reactive simulation with map-based compliance terms. Both are displacement- or compliance-centric: a policy that never reacts to a hazard can score well as long as its trajectory stays close to the recorded one and inside the drivable area. Our Faithfulness axis is designed to be insensitive to exactly the thing these scores reward, and we quantify the disagreement directly rather than asserting it.

b) Causal confusion in imitation learning.: Policies trained by behaviour cloning routinely latch onto correlates of the expert’s action rather than its causes [27]. The standard remedy is intervention at the input; the standard difficulty is constructing an intervention that removes the cause without also removing everything correlated with it. Our three-arm same-frame design, with an equal-area control patch and (for LiDAR consumers) a mirrored-box point deletion, is our answer to that difficulty.

c) Probing and control tasks.: Reading structure out of frozen representations is only meaningful against a control that shares the probe’s capacity [5]. We adopt the random-initialisation control and additionally report a position-only floor, which in our setting accounts for the majority of the raw probe score.

d) Localisation by activation patching.: Replacing an activation with its counterpart from a paired input localises which component carries a behavioural difference. The construction is only interpretable when the patched set is a sufficient cut for that pairing; we enforce this with an explicit patch-all self-check, which in our experiments rejected one policy–corpus cell outright and forced a correction to the intervention itself.

e) Segmentation supervision without labels.: We use class-agnostic masks from SAM [2] as an objective, task-independent probe target, avoiding any human judgement about which objects count as hazards.

III. Method

A. Stimuli

All directions are extracted from the same event pool, mined brake-first: we first locate human decelerations ($\Delta v < -0.5 \text{ m/s}$ with $|\Delta v|/v_0 > 0.3$) and only then attribute them to a front object, so that the presence of a hazard is established by the driver’s behaviour rather than by our own detector. The query frame is the onset of deceleration. Attribution uses a required-deceleration term $a_i = \max(v_0 - v_{\parallel}, 0)^2 / (2 \max(s_i - 2, 0.5))$ inside a 2 m corridor, with four gates (share > 0.6 , $1.5\times$ dominance, explained ratio in $(0.3, 10]$, $a_{\text{req}} \geq 0.4$).

Each event yields three same-frame arms: origin (hazard present), clean (hazard region filled with the median colour of the road surface immediately below it), and ctrl

(an equal-area patch elsewhere). The fill is deliberately noiseless: injecting noise matched to the local texture inflates the measured response four- to eight-fold and contaminates the control arm, which we verified with a three-way comparison and report in Section IV-D.

B. Three directions

Write $h_{\ell}(x)$ for the vision-token-mean activation at layer ℓ .

$$v_{\text{faith}}(\ell) = \frac{1}{n} \sum_i [h_{\ell}(x_i^{\text{orig}}) - h_{\ell}(x_i^{\text{clean}})] \quad (1)$$

$$v_{\text{decel}}(\ell) = \mathbb{E}[h_{\ell} \mid \text{brake}] - \mathbb{E}[h_{\ell} \mid \text{cruise}] \quad (2)$$

$$v_{\text{bright}}(\ell) = \frac{1}{n} \sum_i [h_{\ell}(T(x_i)) - h_{\ell}(x_i)] \quad (3)$$

v_{faith} is paired: both arms are the same frame at the same ego speed, so any common speed component cancels. v_{decel} is a population contrast, stratified into four speed bands so that speed is not confounded with braking. T is a closed-form photometric night transform (gamma 2.2, gain 0.45, saturation 0.45, blue tint, optional Gaussian noise), applied either to the whole image or to a heuristic sky mask; we avoid generative re-rendering because its own artefacts would become a second, uncontrolled domain shift.

C. Reading an angle

In d dimensions two unrelated directions are $\approx 90^\circ$ apart with small variance, so a bare angle is uninterpretable. Every angle we report carries two references: a split-half floor, the median angle between directions fit on two halves of the same data, which is the best agreement the sample size permits; and the random baseline itself. We additionally gate layers on effective dimensionality $\text{PR} = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ of the per-event differences: where $\text{PR} \approx 1$ all perturbations lie on one axis and any angle is forced to 0° or 180° regardless of content. In the policies we study PR rises monotonically with depth ($1.1 \rightarrow 14.5$ for DiffusionDrive), so this gate, together with the floor, selects mid-to-late layers without our having assumed that.

D. Behavioural readout

We read planned speed as full arc length over the horizon, $v^{\text{arc}} = (\|w_1\| + \sum_k \|w_{k+1} - w_k\|) / (N\Delta t)$, rather than a chord: a policy that swerves around a hazard would otherwise be scored as not responding. The appearance effect is $\Delta b = v^{\text{arc}}(T(x)) - v^{\text{arc}}(x)$, positive when the policy plans to go faster after the perturbation. For reference, the hazard-removal response $b = v^{\text{arc}}(x^{\text{clean}}) - v^{\text{arc}}(x^{\text{orig}})$ uses the same sign convention.

E. Pre-registration

Every hypothesis in Section IV-E was written to a timestamped file, with its decision rule and effect-size threshold, before the relevant quantities were computed. This was not ceremony: our first estimate of the leading

TABLE I

Three directions form two families. Angles between the hazard-response direction v_{faith} , the braking direction v_{decel} , and the appearance direction v_{bright} , at each model’s deepest gated layer ($n=312$ left-hand-drive lead events). Read every angle against the split-half floor (what the same data can reproduce) and against 90° (two unrelated directions in high dimensions). v_{faith} and v_{decel} cluster; v_{bright} sits past the random line on the opposite side.

Policy	L^*	angle (deg)			floor		rand
		f-d	b-f	b-d	v_f	v_b	
SimLingo	12	–	133	–	9	10	90
DD	5	44	96	93	5	4	90
LTF	6	46	111	93	5	4	90
DDv2	6	34	111	115	5	5	90

hypothesis showed a perfect rank correlation on four policies, and it fell to $\rho = +0.03$ once two further policies were added. The files are released with the code.

IV. Experiments

A. Setup

Four end-to-end policies: SimLingo (vision–language, 10×0.25 s plans), and the TransFuser family DiffusionDrive, LTF and DiffusionDriveV2 (8×0.5 s; V2 additionally consumes LiDAR). Two further vision–language policies, Alpamayo-R1 and AutoVLA, enter only the pre-registered rank test of Section IV-E; their multi-camera, multi-frame input makes the occlusion intervention cover 1/12 to 1/16 of the visual input rather than all of it, an asymmetry we correct by masking every camera and history frame in which the object projects, and report alongside every number derived from them.

Events come from nuScenes and from the OpenScene/nuPlan data underlying NAVSIM, pooled and then split by driving side: left-hand drive is the benchmark side, right-hand drive the deployment side. The results in Sections IV-B to IV-D are established on the benchmark side ($n=312$ lead events); the deployment side holds 8–10 hazard events and is used only where we say so. Section IV-F explains why that number is a property of the available data rather than of our mining.

B. The three directions form two families

Table I reports the pairwise angles at each policy’s deepest gated layer. v_{faith} and v_{decel} lie $34\text{--}46^\circ$ apart. Against a split-half floor of 7° this is a real separation, and against the 90° random baseline it is a strong association: the two are near but distinct members of one family. This is consistent with what we find in Section IV-E—that v_{faith} carries mostly a braking-tendency component and only weakly a hazard component.

v_{bright} behaves differently. Its angle to both action directions is $93\text{--}115^\circ$, i.e. past the random line, with a mean cosine of -0.54 to v_{faith} across all gated layers and all four policies (39 of 41 layer-cells negative). SimLingo is

TABLE II

Synthetic night makes policies drive faster, and the driver is sensor noise rather than darkness. Δb is the change in planned speed (m/s, v^{arc}) when the same frame is re-rendered as night; positive means faster. “sky” perturbs only the sky region (17.9% of the frame, containing no driving-relevant content); the driving corridor is untouched to the pixel. $n=312$ left-hand-drive lead events.

Policy	sky only		whole image	
	luminance	+noise	luminance	+noise
SimLingo	−0.126	−0.131	+1.485	+1.572
DD	−0.013	+0.045	+0.084	+2.054
LTF	−0.103	+0.091	−0.856	+1.245
DDv2	−0.031	−0.020	−0.116	−0.943

the cleanest case: its cosine sits between -0.67 and -0.76 across all 24 layers, essentially flat with depth.

The sign is what matters. Since $v_{\text{faith}} = h(\text{hazard present}) - h(\text{hazard removed})$, a negative projection says the appearance change moves the representation toward the hazard-removed state. In plain terms, the policy encodes “it got dark” the way it encodes “the hazard is gone.”

C. And the behaviour agrees

Table II gives the behavioural consequence. Under a full-image night rendering with noise, DiffusionDrive plans 2.05 m/s faster, LTF 1.25, SimLingo 1.57. For scale: deleting the lead vehicle from the image raises DiffusionDrive’s planned speed by 0.23 m/s. The appearance perturbation is therefore roughly nine times more effective at making this policy accelerate than the removal of the object it is supposed to be reacting to.

Two independent readouts—the negative cosine of Section IV-B and the positive Δb here—agree on direction, in every policy where both are measurable. DiffusionDriveV2 is the informative exception: it is the only LiDAR-consuming policy, and the only one whose Δb is negative (-0.94). Its representation is also the most tightly coupled ($\cos = -0.84$ at the output layer). Coupling and behavioural sign are separable; the geometry says the appearance axis presses on the action axis, not which way.

a) A single event.: In scene-1055, a Singapore night scene, a pedestrian stands 34 m ahead at the edge of oncoming headlight glare; the human brakes from 9.6 to 3.8 m/s. LTF plans 9.10 m/s and DiffusionDrive 8.58—essentially no deceleration—and removing the pedestrian from the image changes neither ($b = -0.02$ and $+0.02$). The representational and behavioural failures we quantify in aggregate are visible in single frames.

D. Noise, not darkness; global, not local

Two controls separate the mechanism.

a) Luminance versus noise.: Columns 1 and 3 of Table II apply the photometric transform with the noise term set to zero. For DiffusionDrive, sky-only darkening gives $\Delta b = -0.013$ (marginally slower, the safe direction);

adding noise flips it to +0.045. Whole-image: +0.084 without noise, +2.054 with. Darkening is a smooth mono-tone map that preserves edge structure up to a scale factor, and a normalised encoder largely undoes it. Noise adds energy at exactly the spatial frequency where small distant objects live: after darkening, the frame’s own local gradients have $sd \approx 1.9$, and the injected grain has $sd \approx 7$. The evidence is not that darkness is harmless but that the operative variable is sensor noise—which is precisely what a real high-ISO night camera produces. SimLingo is again a partial exception: its whole-image response is large (+1.49) even without noise, so for that policy luminance alone suffices.

b) Sky-only.: The sky mask covers 17.9% of the frame. We verified that the perturbation stays inside it: high-frequency energy rises from 4.29 to 5.85 within the mask and from 6.33 to 6.34 outside, with the with- and without-noise conditions differing by 10^{-4} outside—the residual is feathering, not leakage. The lead vehicle, the road surface and the lane markings are untouched to the pixel. Yet Δb is still +0.045 for DiffusionDrive, 19% of the effect of deleting the hazard entirely. The dependence on appearance is therefore global—plausibly through a pooled or distributional statistic—rather than mediated by any cue in the region that carries the driving decision.

c) Fill colour.: A three-way comparison (grey blob / road colour + noise / road colour only) on 304 deployment lead events shows the road-colour-only condition reproduces the grey blob almost digit for digit (DD +0.234 vs +0.214; LTF −0.153 vs −0.152), while adding noise inflates b four- to eight-fold and drives the control arm from −0.008 to +0.204. All results here use the noiseless road-colour fill.

E. Projections are not a behavioural proxy

Table III lists five pre-registered hypotheses and their outcomes. The pattern is consistent: the projection onto v_{faith} carries some hazard information under favourable conditions—the lead scenario, large-response policies, quartile-level grouping at $n=325$ —and none at the level of individual events, individual policies, or across the domain shift.

Two of the rejections are worth stating precisely, because the manner of failure is informative. The rank hypothesis produced a perfect Spearman correlation on the four policies that generated it and +0.03 once Alpamayo-R1 and AutoVLA were added; AutoVLA has the largest angle and the best closed-loop score, exactly inverted. And the benchmark-to-deployment transfer failed on its own training split (leave-one-out $\delta = 0.02\text{--}0.28$), so the correct reading is not that a direction failed to transfer but that at $n=19$ versus 29 in 256–2048 dimensions there was no direction to begin with.

Table IV shows the one readout that does transfer. Fitting per-layer regression directions against ego speed on benchmark cruising samples, and weighting layers by

TABLE III

Pre-registered tests of the projection as a behavioural proxy, and their outcomes. Every prediction was written to a timestamped file before the relevant numbers existed (results/PREREG_*.md). Four of five are rejected; the fifth holds for two of four policies. We report these because direction readouts are routinely used as behavioural proxies without such checks.

Pre-registered hypothesis	n	Outcome
Occlusion-scenario angle predicts the official closed-loop rank	6 policies	rejected: ρ falls from +1.00 ($n=4$, the set that generated the hypothesis) to +0.03
v_{faith} separates hazard-caused from red-light hard braking	30/49	rejected for 2/4; the other 2 reach $\delta=0.28, 0.30$ against a pre-set bar of 0.33
Representational deviation on the deployment side implies more aggressive behaviour	48/31	rejected: 0/4 policies; three are more conservative
An axis fit on benchmark hard-braking transfers to deployment	on 19/29 $\rightarrow$ 11/20	rejected; the axis fails on its own training split ($\delta=0.02\text{--}0.28$), so no direction exists to transfer
Per-event projection predicts per-event response	325	holds for SimLingo ($\rho=+0.51$) and DDv2 (+0.25); null for DD, LTF

TABLE IV

What does transfer: a multi-layer speed axis. Layer weights and per-layer directions are fit on benchmark cruising samples only; the deployment side is held out entirely. Spearman ρ between the projection and true ego speed. The weighted axis beats the best single layer on held-out data, and the advantage survives the domain shift.

Policy	n		bench (CV)			depl.
	b.	d.	best- L	wtd.	shuf.	wtd.
DD	390	246	+0.65	+0.69	+0.11	+0.62
LTF	390	246	+0.67	+0.72	+0.20	+0.66
DDv2	390	246	+0.57	+0.58	+0.05	+0.52

their held-out agreement with that target, gives $\rho = +0.58$ to +0.72 under 5-fold cross-validation and +0.52 to +0.66 on the held-out deployment side, in every case beating the best single layer and far above a label-shuffled null (+0.05 to +0.20). The contrast with the failures above is instructive: this target variable is external (measured ego speed), whereas v_{faith} ’s target is the policy’s own output.

F. Limits

a) Deployment-side hazard events top out in the low tens.: Four independent criteria converge: brake-first mining yields 74 right-hand-drive events but none with time-to-collision below 2.3s; direct TTC screening yields zero below 1.5s once roadside furniture and cross-traffic are excluded (both verified by inspecting frames); harsh braking ($\geq 3\text{ m/s}^2$) yields 31, of which 11 have a visible object ahead; and the entire NAVSIM right-hand-drive split—1,354 scenes—contains no event above 2 m/s^2 . With

8–10 usable events, a bootstrap over the deployment side gives rank probabilities near chance (27%/40%/33% for one policy across three positions), so we report no deployment-side ranking.

b) The public sensor release is the bottleneck, not the recording.: Deriving per-city hours from log durations—nuPlan publishes no city breakdown—the right-hand-drive city holds 177.3 hours of recorded logs but only 6.9 hours with sensors released (3.9%), against 16.4%, 10.0% and 8.8% for the three left-hand-drive cities.

c) Scope.: Sections IV-B to IV-D are established on the benchmark side. The night transform is closed-form and its noise magnitude is a chosen parameter, so the magnitude of Δb is parameter-dependent; the ordering (noise $\gg$ luminance) and the sign are not. Whether the effect replicates on the deployment side, and whether it is calibrated by real night imagery rather than synthesis, are the two open questions we consider most important.

V. Conclusion

Four end-to-end driving policies share a failure mode we did not set out to find: when a scene is re-rendered as night, the representation moves in a direction systematically opposed to the policy’s own hazard-response direction, and the planned speed rises. The effect is carried by sensor noise rather than by low luminance, and it survives when the perturbation is confined to the sky—a region that contains nothing a driver needs. Two independent readouts, one geometric and one behavioural, agree on this in every policy we tested.

We are less able to say what to do with the directions themselves. Of five pre-registered attempts to use their projections as a behavioural proxy—to rank policies, to flag individual events, to transfer an axis from benchmark to deployment—four are rejected outright, and the one readout that does transfer has an external target variable rather than a model-internal one. The direction that is easiest to extract is not the direction that predicts what the policy will do.

The most severe practical limit is data, not method. Deployment-side hazard events top out in the low tens under four independent mining criteria, and the public sensor release for the right-hand-drive city we studied covers 3.9% of its recorded hours. Any claim that a handful of deployment scenes suffices to estimate a policy’s real behaviour has to contend with that number first.

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Appendix

Appendix

A. Method Changes Relative to the Earlier Protocol

This appendix records superseded definitions so that numbers reported in earlier versions of this work can be located and not confused with the present ones. None of the quantities below appear in the main text.

a) A.1 Event corpus.: The earlier corpus was assembled by a geometric hazard predicate followed by a check that the human braked. Its corridor criterion admitted events in which the entity was already visible in the designated “clean” frame, so the contrast measured a closer approach rather than an appearance. It is fully replaced by the brake-first construction of ??.

b) A.2 Readout.: The earlier readout was the policy’s native `commanded_speed`, covering 0.25–0.75 s of a 2.5–4.0 s plan. On identical events the ground-truth hazard-direction magnitude reads 0.332 under that window and 1.232 over the full horizon, a factor of 3.7. `commanded_speed` is retained only as a sensitivity readout.

c) A.3 Verdict system.: The earlier three-state PASS / FAIL / indeterminate adjudication is replaced by the continuous response ratio of ?? with an interval, which does not require a threshold on a quantity whose scale differs across policies.

A.4 Deployment metrics, written out

The official aggregate combines four multiplicative terms with a weighted sum of four others:

$$\text{PDM} = \left(\prod_{k \in \mathcal{M}} m_k \right) \cdot \frac{\sum_{j \in \mathcal{W}} w_j s_j}{\sum_{j \in \mathcal{W}} w_j}, \quad (4)$$

with $\mathcal{M} = \{\text{no at-fault collision, drivable area, driving direction, traffic light}\}$ and $\mathcal{W} = \{\text{progress, TTC, lane keeping, comfort}\}$. The progress term is normalised within the proposal batch,

$$s_{\text{progress}}^{(i)} = \text{clip} \left(\frac{p_i}{\max_{i'} (p_{i'} \cdot \prod_k m_k^{(i')})}, 0, 1 \right), \quad (5)$$

which is identically 1 when a single proposal is scored alone — the degeneracy reported in ??. We therefore score all candidates as one batch, which makes progress relative to the batch and not comparable with published absolute figures.

Our own comfort readouts are taken from the planned path resampled at $\delta = 0.5$ s, with u_k the speed sequence of ??:

$$a_k = \frac{u_{k+1} - u_k}{\delta}, \quad j_k = \frac{a_{k+1} - a_k}{\delta}, \quad J = \sqrt{\frac{1}{n} \sum_k j_k^2}. \quad (6)$$

B. Corrections Made During This Study

d) B.1 Velocity frame (deployment corpus).: Object velocities in the deployment logs are expressed in the ego frame, not the global frame; an earlier version applied a further rotation, which made every object appear stationary to the attribution of ?? and inflated the lead-vehicle pool by $3.7\times$. The benchmark corpus derives velocity by differencing global positions and was unaffected (verified: pool sizes unchanged).

e) B.2 Turn gate.: A heading-change gate estimated from path tangents produced spurious $\sim 180^\circ$ turns whenever the ego came to a full stop, because the trailing path points coincide. Replacing it with the unwrapped ego-pose yaw plus a 5 m displacement gate removed 33 of 38 events from the affected pool.

f) B.3 Scene identifiers.: Scene names are not unique across dataset splits: 65.8% of the deployment test split’s scene names also occur in the trainval split, referring to different logs. All merged pools are keyed by (split, scene, frame).

g) B.4 Sufficiency of the patched set.: See ??.